Evidence-Aware Risk Scoring for Scientific Claim Verification

Corrected V2 One-Page Academic Summary - 14 October 2026

Scientific claim-verification systems retrieve research papers and decide whether evidence supports or contradicts a claim. These systems can still make confident mistakes: the wrong paper may be retrieved, a decisive sentence may be missed, or a language model may misread the scientific relation. This project asks a second reliability question: can the system identify which of its own predictions are most likely to be wrong?

The project studied Evidence-Aware Risk Scoring (EARS), a selective-prediction approach that ranks claim predictions for possible human review. The matched NLI-only baseline, B1, uses confidence, the support-versus-contradiction margin, and predicted category. EARS adds two label-free evidence-pipeline signals: separation between the top two BM25 retrieval scores and disagreement among evidence-sentence predictions. Both are L2 logistic error rankers trained on the same examples with the same preprocessing and regularization.

The historical frozen Internal Test contained 88 eligible claims in 39 connected groups. EARS improved average risk-coverage ranking (AURC 0.2650 versus 0.3008 for B1) but was worse at the fixed 80% coverage point (R80 0.4571 versus 0.4286). The original joint conclusion therefore remains **H1 INCONCLUSIVE**.

Research Upgrade V2 evaluated official SciFact Dev, a separate split from the same dataset family. Among 188 eligible claims in 164 groups, B1 achieved AURC 0.2816 and R80 0.4133; EARS achieved AURC 0.2884 and R80 0.4000. Paired group-bootstrap intervals crossed zero. The retrieval-margin feature did not improve ranking. The evidence-conflict feature showed exploratory, model-dependent association, but matched ablations did not establish stable incremental benefit. An alternative DeBERTa NLI model also reduced verifier accuracy from 0.4894 to 0.4734.

A final integrity correction found that the original V2 script had deserialized complete Dev JSON records while collecting IDs, although no gold-annotation use in scoring was demonstrated. The correction preserved that history, implemented a restricted ID-and-claim parser, added adversarial label-poisoning tests, and reran both verifiers. Corrected predictions, features, and risk scores were exactly identical to the original V2 scoring artifacts. This verifies engineering equivalence, not a fresh unseen evaluation.

The strongest finding is diagnostic. SUPPORT recall was only 0.323 even though gold-document recall@5 was 0.842; gold-sentence recall fell to 0.516. Finding a relevant paper was not enough to preserve decisive evidence. Shared-document sensitivity also left no stable EARS advantage.

The project therefore offers a careful negative result and a reproducible failure-analysis framework, not a claim of general superiority, external replication, calibrated probabilities, or deployment readiness.

This is a solo-human-owner, substantially AI-assisted project. The human project owner supplied the overarching research objective and integrity constraints, required critical review and retention of mixed results, and retains responsibility for public claims and publication decisions. AI-assisted systems contributed substantially to method and protocol discussions, implementation and debugging, experiment execution, statistical and reproducibility checks, and research writing. Project ownership does not imply that every technical decision or line of code was independently produced without AI. The repository evidence boundary is documented in `docs/PERSONAL_RESEARCH_CONTRIBUTION.md`.